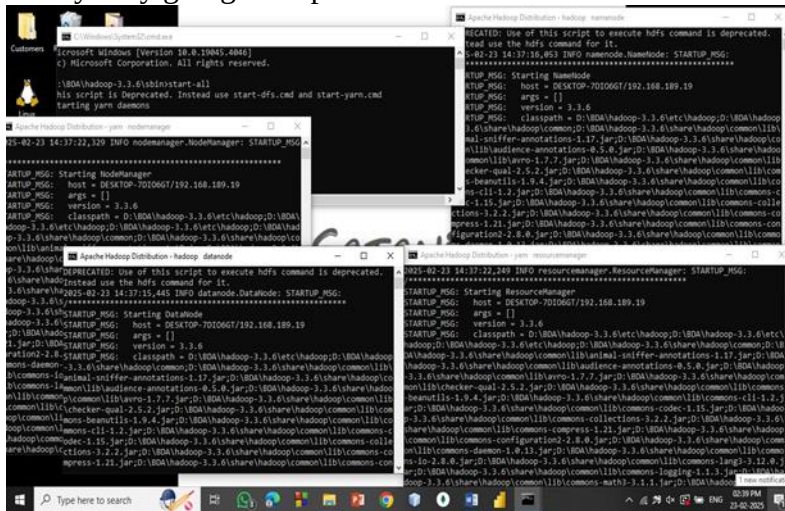


EXPERIMENT-6: Demonstrate pig word count program using Apache Pig.

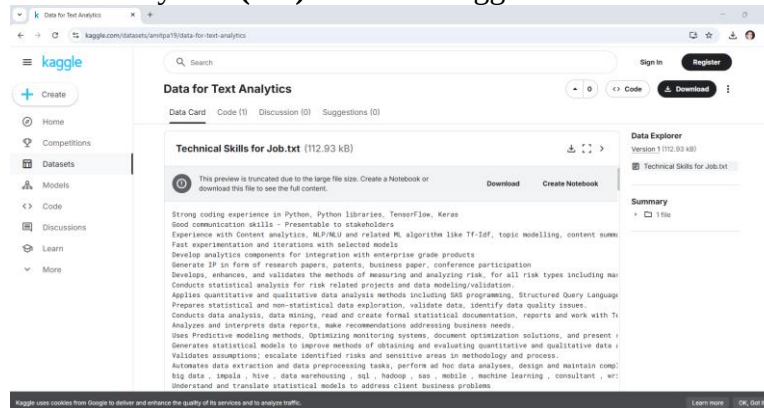
Requirements:

- Start **Hadoop** : Open cmd and run the command start-all
- Verify it by going to <https://localhost:9870> and browse the file system



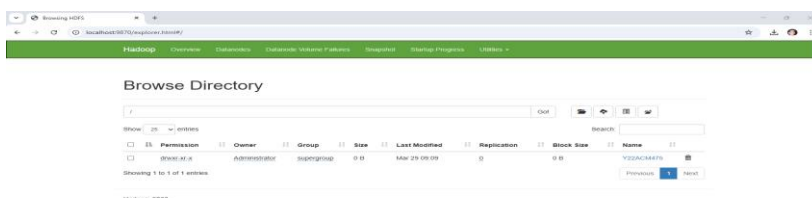
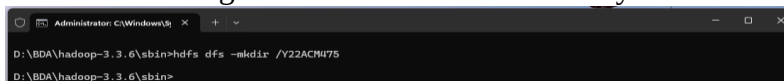
Step1:Download a Text File

- Open Google and search for **Kaggle**.
- Download any **text (.txt)** file from Kaggle.



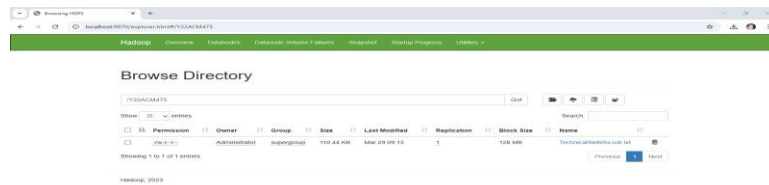
Step 2:Create a Directory in HDFS

- Use the following command to create a directory:



- Move the downloaded file into HDFS using:





Step3: Start Apache Pig

- Launch Pig by running: `pig`
- If successful, the **grunt>** prompt will appear.

```
Administrator: C:\Windows\S...
Microsoft Windows [Version 10.0.22631.4890]
(c) Microsoft Corporation. All rights reserved.

D:\BDA\pig-0.17.0\bin>pig
2025-03-29 09:14:56,421 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2025-03-29 09:14:56,421 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2025-03-29 09:14:56,421 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2025-03-29 09:14:56,490 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2025-03-29 09:14:56,490 [main] INFO org.apache.pig.Main - Logging error messages to: D:\BDA\hadoop-3.3.6\logs\pig_17432
2025-03-29 09:14:56,506 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\Administrator/.pigbootup not found
2025-03-29 09:14:56,658 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2025-03-29 09:14:56,658 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2025-03-29 09:14:56,918 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-default-932f98dc-5254-4e4c-9fc8-a6fadcd8f874
2025-03-29 09:14:56,918 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
grunt>|
```

Step 4: Performing Word Count

Step a: Load Data into Pig

- In the Pig shell, load the data from HDFS:

`Input_lines = LOAD 'Y22ACM475/Technicalskillsforjob.txt' AS (line:Chararray)`

Where, `Input_lines` is called a **relation** in Pig, which acts like a table.

Verify the data is loaded by executing this command: `DUMP Input_Lines;`

```
Administrator: C:\Windows\S...
grunt> Input_lines = LOAD 'Y22ACM475/TechnicalSkillsforJob.txt' AS(line: Chararray);
grunt> dump Input_lines
2025-03-29 09:19:33,361 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: UNKNOWN
2025-03-29 09:19:33,396 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2025-03-29 09:19:33,412 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - [RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter]]
2025-03-29 09:19:33,443 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699480192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2025-03-29 09:19:33,475 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2025-03-29 09:19:33,475 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2025-03-29 09:19:33,475 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2025-03-29 09:19:33,523 [main] INFO org.apache.hadoop.yarn.client.DefaultHadoopFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2025-03-29 09:19:33,612 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2025-03-29 09:19:33,618 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2025-03-29 09:19:33,618 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2025-03-29 09:19:33,618 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2025-03-29 09:19:33,622 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2025-03-29 09:19:33,622 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2025-03-29 09:19:33,622 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.submit.replication is deprecated. Instead, use mapreduce.client.submit.file.replication
2025-03-29 09:19:33,761 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/D:/BDA/pig-0.17.0/pig-0.17.0/pig-0.17.0-core-h2.jar to DistributedCache through /tmp/temp-1656749894/tmp-1538096963/pig-0.17.0-core-h2.jar
2025-03-29 09:19:34,210 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/D:/BDA/pig-0.17.0/pig-0.17.0/lib/automaton-1.11-8.jar to DistributedCache through /tmp/temp-1656749894/tmp-594233175/automaton-1.11-8.jar
2025-03-29 09:19:34,231 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/D:/BDA/pig-0.17.0/pig-0.17.0/lib/antlr-runtime-3.4.jar to DistributedCache through /tmp/temp-1656749894/tmp-899847652/antlr-runtime-3.4.jar
2025-03-29 09:19:34,253 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/D:/BDA/pig-0.17.0/pig-0.17.0/lib/joda-time-2.9.3.jar to DistributedCache through /tmp/temp-1656749894/tmp-1389301884/joda-time-2.9.3.jar
2025-03-29 09:19:34,263 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job
2025-03-29 09:19:34,263 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Key [pig.schematuple] is false, will not generate code.
2025-03-29 09:19:34,263 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Starting process to move generated code to distributed cache
2025-03-29 09:19:34,269 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Setting key [pig.schematuple.classes] with classes to serialize []
2025-03-29 09:19:34,285 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
2025-03-29 09:19:34,285 [JobControl] INFO org.apache.hadoop.yarn.client.DefaultHadoopFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2025-03-29 09:19:34,285 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.task.id is deprecated. Instead, use mapreduce.task.attempt.id
2025-03-29 09:19:34,416 [JobControl] INFO org.apache.hadoop.mapreduce.JobResourceUploader - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Administrator/.staging/job_1743219361106_0001
```

```

umCountWithFixedSleep(mssRetries=10, sleepTime=1000 MILLISECONDS)
2025-03-29 09:27:52.935 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 9 time(s); retry policy is RetryUpToMax
umCountWithFixedSleep(mssRetries=10, sleepTime=1000 MILLISECONDS)
2025-03-29 09:27:55.807 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.MapReduceLauncher - Unable to retrieve job to compute warning aggregation.
2025-03-29 09:27:55.808 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.MapReduceLauncher - [WARNING]
2025-03-29 09:27:55.973 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn
system-metrics-publisher.enabled
2025-03-29 09:27:55.973 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig:schematuple] was not set... will not generate code.
2025-03-29 09:27:55.998 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2025-03-29 09:27:55.998 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapReduceUtil - Total input paths to process : 1
(Strong coding experience in Python, Python libraries, TensorFlow, Hexas)
(Good communication skills - Presentable to stakeholders)
(Experience with Content analysis, ML/PLAI and related ML algorithms like Tf-Idf, topic modelling, content summarization etc)
(Fast experimentation and iterations with selected models)
(Develop analytics components for integration with enterprise grade products)
(Generates IP in form of research papers, patents, business papers, conference participation)
(Develops, enhances, and validates the methods of measuring and analyzing risk, for all risk types including market, credit and operational. Also, may develop, validate and s
strategic uses of scoring models and scoring model related policies.)
(Conducts statistical analysis for risk related projects and data modeling/validation.)
(Applies quantitative and qualitative data analysis methods including SAS programming, Structured Query Language (SQL) to extract, transform and analyze data and Visual Basic
programming languages.)
(Prepares statistical and non-statistical data exploration, validate data, identify data quality issues.)
(Conducts data analysis, data mining, read and create formal statistical documentation, reports and work with Technology to address issues.)
(Analyzes and interprets data reports, make recommendations addressing business needs.)
(Genes Predictive modeling methods, Optimizing monitoring systems, document optimization solutions, and present results to non-technical audiences; write formal documentation
using statistical vocabulary.)
(Generates statistical models to improve methods of obtaining and evaluating quantitative and qualitative data and identify relationships and trends in data and factors affect
ing research results.)
(Validates assumptions; escalate identified risks and sensitive areas in methodology and process.)
(Automates data extraction and data preprocessing tasks, perform ad hoc data analyses, design and maintain complex data manipulation processes, and provide documentation and
presentations.)
(big data, impala, hive, data warehousing, sql, hadoop, sas, mobile, machine learning, consultant, written communication)
(Understand and translate statistical models to address client business problems)
(Lead team in the data synthesis process, solution design and creation of actionable insights, implications and recommendations to address client's business issues in the are
as of marketing optimization, pricing analytics, social/digital analytics.)
(Provide high-end consulting to clients to their business strategy by implementing analytical models using techniques like Linear Regression, Logistic regre
sion, Bayesian regression, Factor analysis, Cluster analysis, Sampling, ARIMA that support the realization of business objectives)
(Design and execute market mix optimization engagement and pricing studies based on historic sales data and develop sophisticated analytical frameworks that add value to the
client and results in new projects and revenue streams.)
(Develop KPI frameworks for evaluating direct and indirect impacts of marketing campaigns.)
(Generate knowledge sharing trends in the team, participate in developing white papers & share new trends in marketing analytics with colleagues and also evolve ones?Ts own o
pinion on the concepts)

```

Step b: Split the Lines into Words

A. Tokenize the Text

- Split each line into individual words:
words = FOREACH raw_data GENERATE FLATTEN(TOKENIZE(line)) AS word;

B. Verify if the Text is Split into Words

- Check if words are separated correctly:DUMP words;

```

grunt Words = FOREACH Input_Lines GENERATE FLATTEN(TOKENIZE(line, ' ')) AS word;
2025-03-29 09:38:04.093 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: UNKNOWN
2025-03-29 09:38:04.099 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn
system-metrics-publisher.enabled
2025-03-29 09:38:04.099 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig:schematuple] was not set... will not generate code.
2025-03-29 09:38:04.099 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - [RULES_ENABLED][addOrderby, ColumnRangePrune, ConstantCalculator, Group
VConstraintAllSettler, ListOptimizer, LoadTypeCastInserter, MapFilter, MapJoinCach, NestedJoinOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushdownVar
EachFlatten, PushJoinFilter, SplitFilter, StreamTypeCastInserter]]
2025-03-29 09:38:04.099 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2025-03-29 09:38:04.099 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2025-03-29 09:38:04.099 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2025-03-29 09:38:04.099 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn
system-metrics-publisher.enabled
2025-03-29 09:38:04.099 [main] INFO org.apache.hadoop.yarn.client.DefaultHadoopRMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2025-03-29 09:38:04.099 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2025-03-29 09:38:04.099 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.JobControlCompiler - Added jar file:/D:/BDA/pig-0.17.0/pig-0.17.0/lib/automa
tom-1.11-0.jar to DistributedCache through /tmp/tmp-1656749894/tmp-18992282/pig-0.17.0-cave-02.jar
2025-03-29 09:38:05.376 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.JobControlCompiler - Added jar file:/D:/BDA/pig-0.17.0/pig-0.17.0/lib/automa
tom-1.11-0.jar to DistributedCache through /tmp/tmp-1656749894/tmp-18992282/pig-0.17.0-cave-02.jar
2025-03-29 09:38:05.392 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.JobControlCompiler - Added jar file:/D:/BDA/pig-0.17.0/pig-0.17.0/lib/automa
tom-1.11-0.jar to DistributedCache through /tmp/tmp-1656749894/tmp-18992282/pig-0.17.0-cave-02.jar
2025-03-29 09:38:05.826 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.JobControlCompiler - Added jar file:/D:/BDA/pig-0.17.0/pig-0.17.0/lib/joda-t
ime-2.9.3.jar to DistributedCache through /tmp/tmp-1656749894/tmp-18992282/pig-0.17.0-cave-02.jar
2025-03-29 09:38:05.826 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.JobControlCompiler - Setting up single store job
2025-03-29 09:38:05.826 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Key [pig:schematuple] is false, will not generate code
2025-03-29 09:38:05.826 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Starting process to move generated code to distributed cache
2025-03-29 09:38:05.826 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Setting key [pig:schematuple.class] with classes to deserialize []
2025-03-29 09:38:05.832 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
2025-03-29 09:38:05.832 [JobControl] INFO org.apache.hadoop.yarn.client.DefaultHadoopRMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2025-03-29 09:38:05.832 [JobControl] INFO org.apache.hadoop.mapreduce.JobResourceUploader - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Administrator/.stagin
g/job_YU22ACM475/186.9807
2025-03-29 09:38:05.853 [JobControl] WARN org.apache.hadoop.mapreduce.JobResourceUploader - No job jar file set. User classes may not be found. See Job or JobSetJar(String
)
2025-03-29 09:38:05.853 [JobControl] INFO org.apache.builtin.PigStorage - Using PigTextInputFormat
2025-03-29 09:38:05.853 [JobControl] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2025-03-29 09:38:05.853 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapReduceUtil - Total input paths to process : 1
2025-03-29 09:38:05.853 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapReduceUtil - Total input paths (combined) to process : 1
2025-03-29 09:38:06.296 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - number of splits:1

```

```

(Notable)
(action)
(through)
(fact-based.)
(inspirational)
(material.)
(hus)
(a)
(track)
(record)
(er)
(identifying)
(ind)
(championing)
(new)
(processes)
(fur)
(improving)
(fact-based)
(decision)
(making.)
(can)
(synthesize)
(multiple.)
(disparate)
(data)
(sources)
(into)
(compelling)
(growth)
(strategies.)
(Formulates)
(a)
(strong)
(POV)
(hus)
(can)
(particulate)
(future)
(scenarios;)
(To)
(an)

```

Step c: Group Words

A. Group All Occurrences of Each Word

- Organize words into groups based on their value:
grouped_words = GROUP words BY word;

B. Verify if Words are Grouped Properly

- Display grouped words to check correctness:
DUMP grouped words;

```

hadoop> Grouped = GROUP Words BY word;

2025-03-29 09:39:36.576 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY
2025-03-29 09:39:36.592 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn
system-metrics-publisher.enabled
2025-03-29 09:39:36.592 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2025-03-29 09:39:36.592 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupB
yConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownFor
EachFilter, PushUpFilter, SplitFilter, StreamTypeCastInserter]}
2025-03-29 09:39:36.592 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCCompiler - File concatenation threshold: 180 optimistic? false
2025-03-29 09:39:36.592 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2025-03-29 09:39:36.592 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2025-03-29 09:39:36.602 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn
system-metrics-publisher.enabled
2025-03-29 09:39:36.607 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMAFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2025-03-29 09:39:36.607 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set,
set to default 0.3
2025-03-29 09:39:36.607 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducer
s
2025-03-29 09:39:36.612 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.hadoop.
executionengine.mapReduceLayer.InputSizeReducerEstimator
2025-03-29 09:39:36.612 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator - BytesPerReducer:1000000000 maxReducers:999 total
InputFileSize:113094
2025-03-29 09:39:36.612 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 1
2025-03-29 09:39:36.612 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2025-03-29 09:39:36.639 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/D:/BDA/pig-0.17.0/pig-0.17.0
-core-h2.jar to DistributedCache through /tmp/temp-1656749894/tmp-1347786931/pig-0.17.0-core-h2.jar
2025-03-29 09:39:36.640 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/D:/BDA/pig-0.17.0/pig-0.17.0/lib/automa
ton-1.11-8.jar to DistributedCache through /tmp/temp-1656749894/tmp-704277754/automa-1.11-8.jar
2025-03-29 09:39:36.685 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/D:/BDA/pig-0.17.0/pig-0.17.0/lib/anti-r
untime-3.4.jar to DistributedCache through /tmp/temp-1656749894/tmp-1862382288/anti-r-3.4.jar
2025-03-29 09:39:37.116 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/D:/BDA/pig-0.17.0/pig-0.17.0/lib/joda-t
ime-2.9.3.jar to DistributedCache through /tmp/temp-1656749894/tmp-19812670/joda-time-2.9.3.jar
2025-03-29 09:39:37.116 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job
2025-03-29 09:39:37.116 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Key [pig.schematuple] is false, will not generate code
2025-03-29 09:39:37.116 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Starting process to move generated code to distributed cache
2025-03-29 09:39:37.116 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Setting key [pig.schematuple.classes] with classes to deserialize []
2025-03-29 09:39:37.133 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
2025-03-29 09:39:37.133 [JobControl] INFO org.apache.hadoop.yarn.client.DefaultNoHARMAFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2025-03-29 09:39:37.150 [JobControl] INFO org.apache.hadoop.mapreduce.JobResourceUploader - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Administrator/.stagin
g/job_1703219361106_0003

```

```

(learning/description, ((learning/description)))
(advantages/drawbacks, ((advantages/drawbacks, (advantages/drawbacks)))
(algorithms/experience, ((algorithms/experience)))
(analytics/statistical, ((analytics/statistical)))
(cross-regional/global, ((cross-regional/global)))
(preparing/integrating, ((preparing/integrating)))
(technology/operations, ((technology/operations)))
(management/statistical, ((management/statistical, (management/statistical)))
(predictive/descriptive, ((predictive/descriptive)))
(processing/performance, ((processing/performance)))
(tableau/microstrategy, ((tableau/microstrategy, (tableau/microstrategy)))
(developing/configuring, ((developing/configuring)))
(education/professional, ((education/professional)))
(relationship-marketing, ((relationship-marketing)))
(visualizing/presenting, ((visualizing/presenting, (visualizing/presenting)))
(2e/3e-tech/web-tech/mba, ((2e/3e-tech/web-tech/mba)))
(certification/trainings, ((certification/trainings)))
(intelligence/analytics, ((intelligence/analytics)))
(optimization/operations, ((optimization/operations, (optimization/operations)))
(ridge-lasso-elasticnet, ((ridge-lasso-elasticnet)))
(slack/jira/asana/trello, ((slack/jira/asana/trello, (slack/jira/asana/trello)))
(organization/functional, ((organization/functional)))
(predictive/prescriptive, ((predictive/prescriptive, (predictive/prescriptive)))
(intelligence/informatics, ((intelligence/informatics)))
(modeling/sql/python/data, ((modeling/sql/python/data, (modeling/sql/python/data)))
(advanced/matrices, ((advanced/matrices)))
(geographically-dispersed, ((geographically-dispersed)))
(labeling/categorization, ((labeling/categorization)))
(segmentation/clustering, ((segmentation/clustering)))
(sensitivities/corridors, ((sensitivities/corridors, (sensitivities/corridors)))
(tableau/qlik/gplot/shiny, ((tableau/qlik/gplot/shiny)))
(campaigns/programs/tactics, ((campaigns/programs/tactics)))
(conferences/qualifications, ((conferences/qualifications)))
(representatives/stakeholders, ((representatives/stakeholders)))
(research/insights/analytics, ((research/insights/analytics)))
(management, ((management, (management, detailing/demand)))
(visualization/statistics/model, ((visualization/statistics/model, (visualization/statistics/model)))
(lending/liabilities/investment, ((lending/liabilities/investment)))
(iiits/aiiits/aiit/iiitms/psu), ((iiits/aiiits/aiit/iiitms/psu)))
((mathematics/statistics/financial, ((mathematics/statistics/financial)))
((predictivemodelsneuralnetworks#deeplearning#advanced, ((predictivemodelsneuralnetworks#deeplearning#advanced)))
(learning#technique#model#algorithm#spark#mapreduce#tensorflow#artificialintelligence#nlp, ((learning#technique#model#algorithm#spark#mapreduce#tensorflow#artificial

```

Step d: Count Word Occurrences

A. Count the Number of Times Each Word Appears

- Use Pig's built-in function to count occurrences:
word_count = FOREACH grouped_words GENERATE group AS
word,COUNT(words) AS count;

B. Verify the Word Count

- Check if words are counted correctly:
DUMP word_count;


```

Administrator C:\Windows\S...
print> Wordcount = FOREACH Grouped GENERATE group, COUNT(Word);
print> DUMP Wordcount;
2025-03-29 09:49:00,736 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig Features used in the script: GROUP_BY
2025-03-29 09:49:00,736 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn
system-metrics-publisher.enabled
2025-03-29 09:49:00,736 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2025-03-29 09:49:00,736 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2025-03-29 09:49:00,736 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - [RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupB
yConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownFor
EachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter]]
2025-03-29 09:49:00,736 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2025-03-29 09:49:00,751 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2025-03-29 09:49:00,751 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2025-03-29 09:49:00,751 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn
system-metrics-publisher.enabled
2025-03-29 09:49:00,751 [main] INFO org.apache.hadoop.yarn.client.DefaultHadoopMRFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2025-03-29 09:49:00,751 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2025-03-29 09:49:00,751 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set,
set to default 0.3
2025-03-29 09:49:00,751 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducer
s.
2025-03-29 09:49:00,751 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.hadoop.
executionengine.mapReduceLayer.InputSizeReducerEstimator
2025-03-29 09:49:00,761 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator - BytesPerReducer=1000000000 maxReducers=999 total
InputFileSize=113094
2025-03-29 09:49:00,761 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 1
2025-03-29 09:49:00,761 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2025-03-29 09:49:00,767 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/D:/BDA/pig-0.17.0/pig-0.17.0/lib/0.17.0
-core-h2.jar to DistributedCache through /tmp/temp-1656749894/tmp-1251416229/pig-0.17.0-core-h2.jar
2025-03-29 09:49:00,799 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/D:/BDA/pig-0.17.0/pig-0.17.0/lib/automa
ton-1.11-8.jar to DistributedCache through /tmp/temp-1656749894/tmp1990818336/automa-ton-1.11-8.jar
2025-03-29 09:49:00,829 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/D:/BDA/pig-0.17.0/pig-0.17.0/lib/antlr-
runtime-3.4.jar to DistributedCache through /tmp/temp-1656749894/tmp-926423886/antlr-runtime-3.4.jar
2025-03-29 09:49:01,271 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/D:/BDA/pig-0.17.0/pig-0.17.0/lib/joda-t
ime-2.9.3.jar to DistributedCache through /tmp/temp-1656749894/tmp-1637271323/joda-time-2.9.3.jar
2025-03-29 09:49:01,277 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job
2025-03-29 09:49:01,277 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Key [pig.schematuple] is false, will not generate code.
2025-03-29 09:49:01,277 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Starting process to move generated code to distributed cacche
2025-03-29 09:49:01,277 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Setting key [pig.schematuple.classes] with classes to deserialize []
2025-03-29 09:49:01,281 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
2025-03-29 09:49:01,281 [JobControl] INFO org.apache.hadoop.yarn.client.DefaultHadoopMRFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2025-03-29 09:49:01,303 [JobControl] INFO org.apache.hadoop.mapreduce.JobResourcePlanner - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Administrator/.stagin

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Administrator C:\Windows\S...
(solutions/approach,1)
(tests/experiments,1)
(Analytics/Marketing,1)
(Dashboard/Reporting,1)
(FMCG/Pharmaceutical,1)
(Scipy,Numpy,Pandas,1)
(analysis/supervised,1)
(client-relationship,1)
(high-dimensionality,1)
(products/structures,1)
(results/conclusions,2)
(Skills/experiences,1)
(Management,Marketing,1)
(input-process-output,1)
(knowledge/processes,1)
(modeling/validation,1)
(structure/packaging,1)
(website/app/internet,1)
(Learning,Presentation,1)
(LearningDescriptionIf,1)
(Advantages/Drawbacks,2)
(Algorithms,Experience,1)
(Analytics/statistical,1)
(cross-regional/global,1)
(preparing/integrating,1)
(technology/operations,1)
(Management,Statistical,2)
(Predictive/Descriptive,1)
(Processing/Performance,1)
(Tableau/Microstrategy,2)
(developing/configuring,1)
(education/professional,1)
(relationship-marketing,1)
(visualizing/presenting,2)
(BE/B-Tech/ME/M-Tech/MBA,1)
(Certification/Trainings,1)
(Intelligence/Analytics,1)
(Optimization/Operations,2)
(Ridge-Lasso-ElasticNet,1)
(Slack/JIRA/Asana/Trello,2)
(organization/functional,1)
(predictive/prescriptive,2)

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